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Data Science- Assignment 3_Regression_Baseline_Diagnostics

Dataset: scikit-learn diabetes regression benchmark

Analysis design: Frozen 80/20 split; CV tuning on training data only

Github: https://github.com/amsalu2212/DS-Ass3/blob/main/DS_A3_RegressionBlDiagnosis.ipynb

Result summary

This report establishes a transparent regression baseline and examines where its predictions are reliable or weak. The analysis uses the bundled diabetes regression benchmark from scikit-learn (442 rows and 10 features). The target is a quantitative disease-progression measure in the benchmark, not a diagnosis or a clinical recommendation. A fixed random split assigned 353 records to training and 89 to testing. Preprocessing was placed inside model pipelines, and Ridge/Lasso hyperparameters were selected with cross-validation using training data only. On the held-out test set, CV-tuned Lasso had the lowest RMSE (53.992) and highest R^2 (0.518); Ridge and ordinary linear regression were close behind. Relative to the mean-only baseline, Lasso reduced RMSE by 30.8% on this split. The improvement is meaningful as a baseline comparison, but the small gap among the three feature-based models suggests that regularization provides only a modest incremental gain over ordinary least squares here.

Diagnostics qualify the headline score. Test residuals ranged from −123.618 to +107.586, with a mean of −3.048. The Breusch–Pagan test did not provide evidence of heteroscedasticity at the conventional 0.05 level (LM p=0.266; F p=0.268), but this is not proof of constant variance. Errors also varied across BMI slices: the highest training-defined BMI quartile had RMSE 66.187 compared with 37.981 in the lowest quartile. The defensible conclusion is therefore that regularized linear models outperform a constant prediction on this fixed benchmark split, while substantial individual errors and subgroup variation remain.

Main results at a glance

Best test model	Lasso: RMSE 53.992; MAE 42.989; R^2 0.518	Best among the candidates on this fixed test split; not proof of external generalization.
Baseline improvement	RMSE falls from 77.976 to 53.992 (30.8%)	Feature information improves prediction over always predicting the training mean.
Regularization gain	Lasso RMSE 53.992 vs linear 54.281	Small gain; report as modest rather than transformative.
Uneven error slices	BMI quartile RMSE: 37.981 to 66.187	Performance is not uniform across this test-set slice analysis.
Leakage controls	353 train / 89 test; split disjoint; tuning train-only	Supports a fair internal comparison when the notebook is rerun as documented.

1. Objective and research questions

The objective is to establish an interpretable baseline and use residual and error slices to define its boundary. The work addresses five questions as follows:

1) Does a feature-based model improve on a naive mean predictor?
2) D does regularization improve held-out performance?
3) What do residual and influence diagnostics reveal?
4) Does performance vary across a meaningful slice? and
5) What claims are justified by the observed evidence?

A logistic model was not fitted because the selected target is continuous and the assignment is a regression task. Logistic regression is generally used for categorical outcomes; converting the benchmark target to a class would change the task rather than provide a like-for-like regression candidate. The candidates are therefore a mean baseline, ordinary linear regression, Ridge, and Lasso.

2. Dataset and preparation

The notebook loads the diabetes benchmark directly with `sklearn.datasets.load_diabetes(as_frame=True, scaled=True)`. It contains 442 observations and 10 named features (`age`, `sex`, `bmi`, `bp`, `s1`–`s6`) with no missing feature values. The features are already standardized by the dataset provider; the modeling pipeline also places StandardScaler within the training pipeline so that any fitted preprocessing remains confined to training folds. No external CSV or internet download is needed.

This dataset is a compact teaching benchmark, which makes the analysis reproducible and fast. Its size and provenance also limit the scope of inference: performance on it should not be generalized to current clinical populations, other institutions, or individual medical decisions. The target is used as a numerical benchmark outcome only.

3. Split strategy and leakage audit

The split was frozen using `train_test_split(test_size=0.20, random_state=20261015)`. This produced 353 training rows and 89 test rows. The test partition was reserved for final reporting. Five-fold cross-validation selected Ridge and Lasso regularization strengths using the training partition; the held-out test scores were not used to choose hyperparameters.

Table I: Split strategy and leakage audit with hyper parameter tuning

Train/test indices disjoint	PASS intersection empty	Prevents the same row appearing in both partitions.
All rows accounted for	PASS - 353 + 89 = 442	Confirms the split covers the complete benchmark.
Preprocessing inside pipeline	PASS - scaler fitted within pipelines/CV folds	Prevents information from validation folds leaking into preprocessing.
Tuning on training data only	PASS - GridSearchCV fit on training partition	Keeps test set independent of hyperparameter selection.
Test set used for model selection	NO	Preserves test set as final evaluation evidence.

These checks support the intended leakage-free design, assuming the notebook is run as written and the split is not later changed in response to test performance. The fixed seed improves reproducibility; it does not remove sampling uncertainty associated with a single split.

4. Models and evaluation metrics

The naive baseline predicts the mean target value from the training set for every test case. Ordinary linear regression estimates a linear relationship between the features and target. Ridge adds an L2 penalty that shrinks coefficients, while Lasso adds an L1 penalty that can shrink some coefficients to exactly zero. These penalties can reduce variance and simplify the fitted model, especially when features are correlated.

RMSE gives larger errors more influence because errors are squared before averaging; MAE describes average absolute error in target units and is less sensitive to very large errors than RMSE. R^2 measures improvement relative to predicting the test-set mean; values nearer 1 indicate stronger predictive fit, while a negative value indicates worse performance than a constant test-mean predictor. The three metrics should be read together.

5. Held-out model comparison

Table II: different model baselines for the proposed data analysis

Model	Test RMSE	Test MAE	Test R^2	Alpha	Train CV RMSE
Lasso (CV-tuned)	53.992	42.989	0.518	0.50	54.882
Ridge (CV-tuned)	54.207	42.891	0.514	1.00	54.932
Linear regression	54.281	42.918	0.513	—	—
Dummy mean baseline	77.976	65.799	-0.005	—	—

The mean baseline is clearly weakest (RMSE 77.976; MAE 65.799; R^2 −0.005), confirming that the available features carry useful predictive information. Lasso has the best RMSE (53.992) and R^2 (0.518), while Ridge has the lowest MAE by a small margin (42.891 versus Lasso's 42.989). This metric-specific difference matters: there is no single metric-independent winner. If large misses are the main concern, RMSE favors Lasso; if average absolute deviation is the priority, Ridge is marginally better.

Lasso's RMSE is about 0.289 lower than ordinary linear regression (53.992 versus 54.281), a small improvement on this split. Ridge is also close (54.207). The tuned alpha values were 0.5 for Lasso and 1.0 for Ridge, selected by training-only cross-validation. These results suggest regularization is reasonable, but the observed gain over ordinary least squares is modest and should not be oversold.

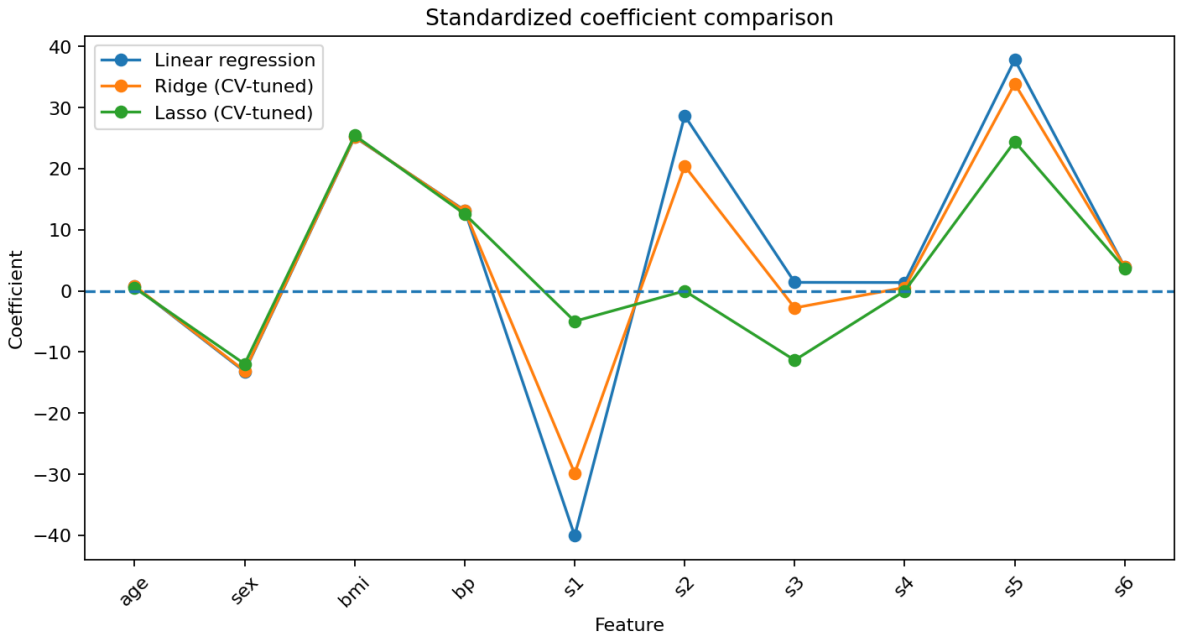


Figure 1. Standardized coefficient comparison across ordinary linear regression, Ridge, and Lasso. Coefficients are fitted associations, not causal effects.

Figure 1 shows how coefficient estimates change under regularization. Ordinary least squares has comparatively large, opposing coefficients for s1 and s2, a pattern that can occur when predictors are correlated. Ridge shrinks coefficients toward zero more smoothly; Lasso shrinks more aggressively and sets some coefficients (including s2 and s4 in this fit) to zero. This improves compactness and can aid stability, but a zeroed coefficient is not proof that the underlying feature is scientifically irrelevant.

6. Residual diagnostics

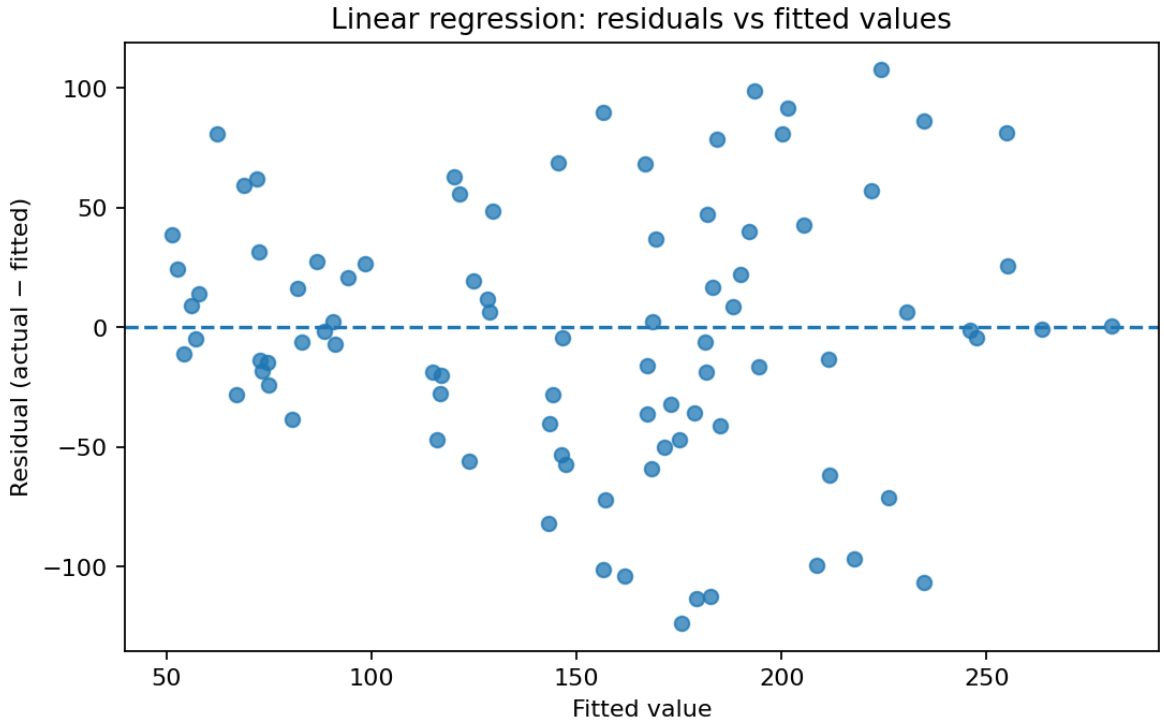


Figure 2. Test residuals against fitted values for ordinary linear regression. Residual = actual − predicted; the dashed line marks zero error.

The test residual mean is -3.048, close to zero, which is expected when positive and negative errors offset. The residual standard deviation is 54.502; observed residuals range from -123.618 to 107.586. Figure 2 is used to look for systematic curvature, changing spread, or clusters rather than to judge the model from a single summary statistic. The plot should be inspected for structure; any visual pattern is diagnostic evidence, not proof by itself.

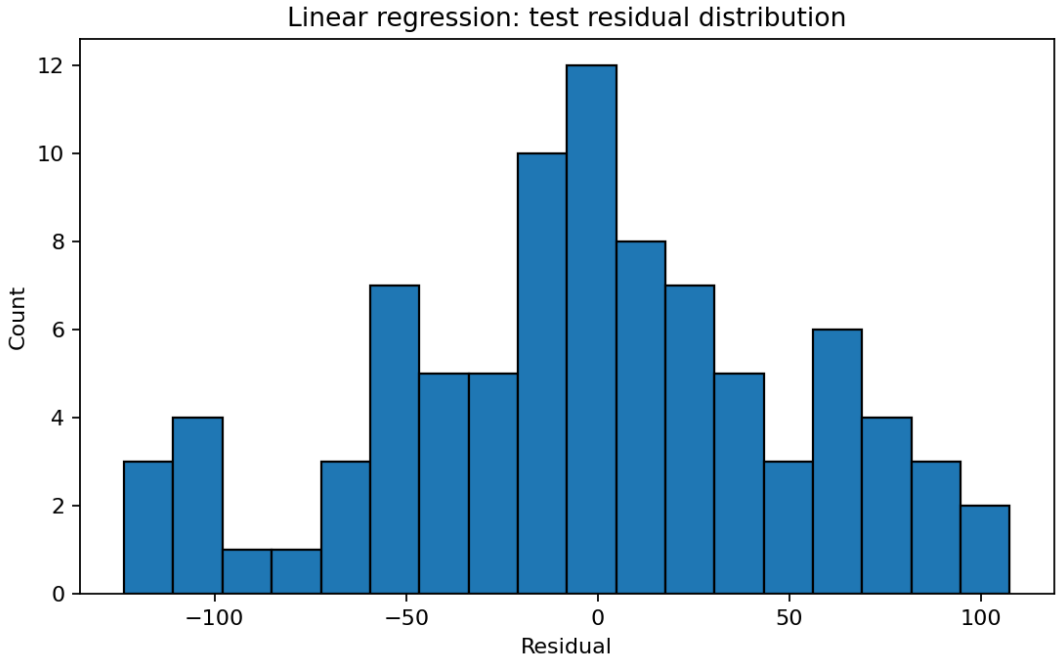


Figure 3. Distribution of held-out test residuals for ordinary linear regression.

Figure 3 summarizes the balance and spread of the errors. A broad distribution and long tails are consistent with the large individual misses reported below. Residuals need not be perfectly normal for predictions to be useful, but strong skew or tail behavior can affect uncertainty statements and highlights the importance of reporting MAE/RMSE alongside plots. These figures are diagnostics for the linear model, not a guarantee that the same residual behavior applies to every candidate or to future populations.

7. Influence and heteroscedasticity

Influence statistics were calculated on the training partition using an OLS model fitted to standardized training features. Studentized residuals indicate unusually large errors relative to an estimated residual scale; leverage indicates unusual feature-space positions; Cook's distance summarizes how much a case may influence the fitted regression. These flags identify cases for review, not cases to delete automatically.

Table III: . Influence statistics with heteroscedasticity

Training row index	Studentized residual	Leverage	Cook's distance
382	-2.309	0.069	0.0357
123	-1.708	0.088	0.0255
58	1.483	0.108	0.0242
304	2.489	0.041	0.0240
387	2.063	0.057	0.0231
169	-1.233	0.142	0.0228

The largest Cook's distance in the displayed training table is 0.0357 (row index 382). Its value is a prompt to check data integrity and model sensitivity; it is not, by itself, a reason to exclude that observation. No claim is made that these cases are errors in the source data.

The Breusch–Pagan test returned LM statistic 12.284 with p=0.266, and F statistic 1.233 with p=0.268. Both p-values exceed 0.05, so this test does not provide statistically significant evidence against constant residual variance in the training OLS fit. Failure to reject the null is not proof of homoscedasticity; the plot, test power, sample size, and model specification all matter. This test assesses a model assumption, not predictive accuracy.

8. Learning curve and data sufficiency

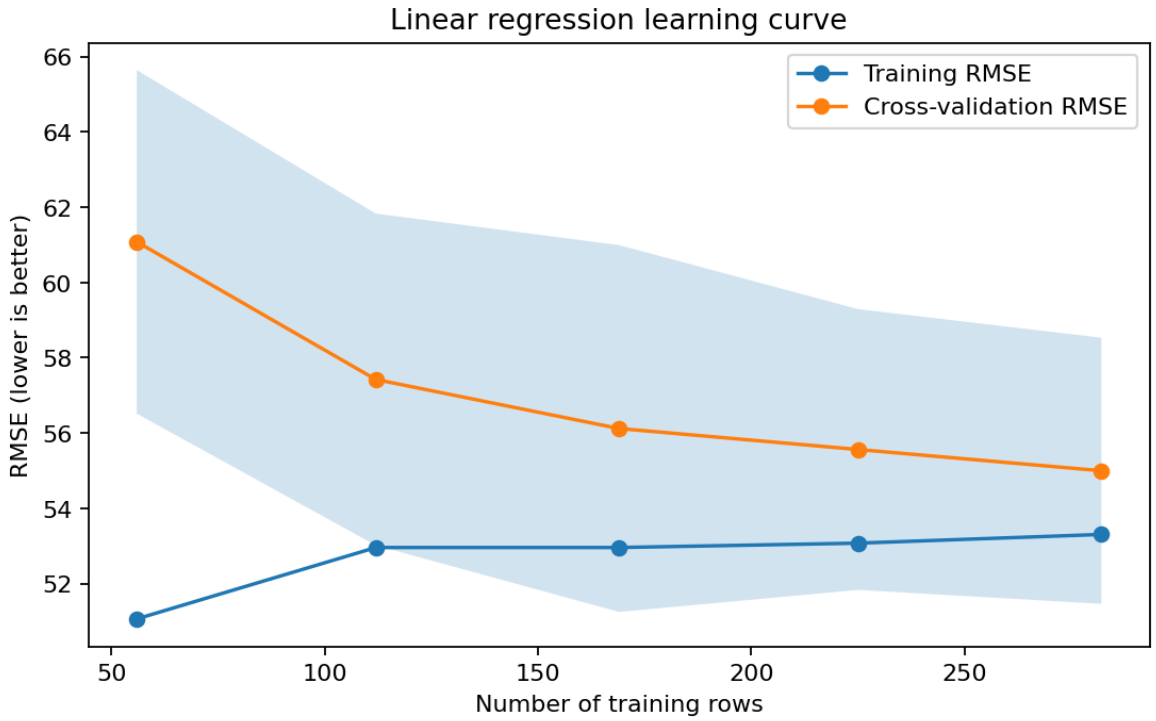


Figure 4. Training and cross-validation RMSE for ordinary linear regression as the number of training examples increases.

The learning curve shows mean training RMSE rising from 51.063 at 56 training rows to 53.307 at 282 rows, while cross-validation RMSE falls from 61.079 to 54.999. The narrowing gap is consistent with better generalization as more examples are used and suggests that the smallest training subsets are unstable. The validation curve is still improving at the largest evaluated size, so additional representative data could help; this curve alone cannot quantify how much improvement a larger dataset would deliver.

9. Critical slice analysis and individual errors

To check whether performance is uniform across a meaningful feature range, BMI quartile cut points were calculated from the training BMI values only and then applied to the test partition. This avoids defining slice boundaries from the held-out outcomes. The slice results are descriptive and the sample sizes are small, so they should be treated as signals for further investigation rather than definitive subgroup rankings.

Table IV: Critical slice analysis and individual errors across different metrics

Test BMI slice	n	RMSE	MAE	R^2
BMI: lowest training quartile	24	37.981	31.444	0.072
BMI: second quartile	17	50.946	37.644	0.176
BMI: third quartile	21	55.723	47.206	0.424
BMI: highest training quartile	27	66.187	53.103	0.396

The highest BMI quartile has the largest RMSE (66.187) and MAE (53.103), while the lowest quartile has RMSE 37.981 and MAE 31.444. This gap indicates that aggregate metrics mask differences across the feature range. Possible explanations include nonlinearity, different target variability, limited sample size, or interactions not captured by a simple linear form; the analysis does not establish which explanation is correct. The appropriate next step is to investigate model form and evaluate the slice pattern on additional data, without tuning on this test set.

Table V: Performance metrics for differences across the feature range

Test row index	Actual	Linear prediction	Residual	Absolute error
380	52	175.618	-123.618	123.618
274	66	179.302	-113.302	113.302
211	70	182.672	-112.672	112.672
290	332	224.414	107.586	107.586
147	128	234.741	-106.741	106.741

The largest absolute error is 123.618 (actual 52, predicted 175.618), and the next two errors exceed 112 points in absolute value. Other cases are overpredictions as well as underpredictions, so errors are not exclusively in one direction. Such cases are important evidence against claiming uniformly accurate predictions. They should be examined for data issues and possible missing structure, but not removed simply because they worsen the score.

10. Feature importance and interpretation boundary

Permutation importance estimates how much a selected model's predictive score deteriorates when one feature is shuffled in the held-out data. In this run, s1 has the largest mean importance (25.621 RMSE units), followed by s5 (20.530), BMI (11.476), s2 (11.044), and blood pressure (4.642). The mean importance for age is slightly negative (−0.053), which can occur from sampling variation or correlated predictors; it should be read as no clear positive importance signal under this specific procedure, not as evidence that age never matters.

Table VI: Feature importance and interpretation boundary

Feature	Mean permutation importance	Std. across repeats
s1	25.621	4.490
s5	20.530	2.909
bmi	11.476	1.977
s2	11.044	2.658
bp	4.642	1.274
sex	0.823	0.845

Permutation importance is model- and sample-dependent. With correlated features, shuffling one variable may understate its contribution because other variables contain overlapping information. Likewise, standardized regression coefficients can be unstable when predictors are correlated. Neither coefficient magnitude nor permutation importance supports causal claims; the analysis is predictive, not an intervention study.

11. Conclusions and defensible claims

- On the fixed 89-row test set, Lasso achieved the lowest RMSE (53.992) and highest R^2 (0.518) among the four candidates; Ridge had the lowest MAE by a small margin.
- Compared with the mean baseline, Lasso reduced test RMSE by 30.8% on this one split, showing that feature-based linear models add predictive value for this benchmark.
- The incremental RMSE improvement of Lasso over ordinary linear regression was small, so the evidence supports modest rather than dramatic gains from regularization.
- Residual ranges, large individual errors, and the BMI slice gap show that errors remain substantial and are not uniform across all cases.
- The Breusch–Pagan test did not detect significant heteroscedasticity at 0.05, but this is not proof of constant variance. Influence flags should prompt review rather than automatic exclusion.
- Coefficients and permutation importance describe predictive associations in this model and data; they do not show that changing a feature would cause a change in the target.

Limitations

- The dataset is small and intended as a teaching benchmark; external validity is limited.
- Only one fixed train/test split is reported. Repeated splits or external validation would better characterize performance variability.
- The models are linear in the provided features; nonlinearities and interactions were not exhaustively searched.
- Slice analysis is exploratory and the slice sample sizes are modest.
- This is not a clinical prediction system and must not be used for diagnosis or treatment decisions.

12. Reproduction and submission checklist

- Open `DS-A3_Regression_Baseline_Diagnostics.ipynb` in Jupyter Notebook, JupyterLab, or VS Code.
- Install compatible versions of numpy, pandas, matplotlib, scikit-learn, statsmodels, and IPython.
- Select Restart Kernel and Run All. The notebook loads its benchmark dataset from scikit-learn and writes outputs to `ds_a3_outputs/`.
- Keep the notebook, generated evidence tables and figures, dependency/environment record, and AI-use log together in version control.
- Verify whether the instructor requires continuity with the dataset used in A1/A2 before final submission.
- Record the repository URL and commit/tag used for the platform submission; do not claim independent authorship of AI-assisted work.

AI-use and integrity note

- AI assistance was used to help structure the analysis and prepare the explanatory report. Before submission, the student should complete the course's AI-use log with the actual tool/tasks used, advice accepted or rejected, and the checks performed independently. The report summarizes recorded notebook outputs; it should be rerun and checked by the student in their own environment. Do not alter or invent results to fit a desired conclusion.

Appendix: metric definitions

Table VII: Model performance metrics

RMSE	Square root of mean squared prediction error	Lower is better; disproportionately penalizes large errors.
MAE	Mean absolute prediction error	Lower is better; average magnitude of error in target units.
R^2	1 − residual sum of squares / total sum of squares	Higher is better; negative values can occur when worse than a constant benchmark.
Cook's distance	Influence diagnostic combining residual size and leverage	Large values flag cases to review; not automatic grounds for deletion.
Breusch–Pagan p-value	Test of whether residual variance relates to predictors	Small p-value is evidence against constant variance; large p-value does not prove it.
Permutation importance	Score decrease after permuting a feature	Model- and dataset-dependent; not causal importance.